



From Monolith to Modular Fabric: Securing the Next Generation of Multi- Agent Systems with a Layered Protocol Architecture

The evolution of artificial intelligence is increasingly defined by the transition from single, general-purpose models to complex ecosystems of specialized AI agents. This shift presents a fundamental challenge not just in tool invocation, but in enabling these diverse entities to discover one another, advertise capabilities, coordinate tasks, and communicate in a standardized, secure, and observable manner . The Agent Fabric Design Memo proposes a comprehensive architectural framework designed to address this challenge directly, moving beyond the limitations of existing protocols like the Model Context Protocol (MCP). This report provides a deep analysis of the fabric's design principles, its layered protocol structure, its security model, and its potential to foster broad adoption across the AI ecosystem. The analysis evaluates how the proposed architecture, which deliberately separates concerns through distinct layers for discovery, capability exposure, and coordination, can create a more robust and trustworthy environment for multi-agent collaboration. It specifically examines the role of the structured event fabric, the innovative DMZ watcher pattern for security, and the pragmatic use of established network standards to build a scalable and interoperable foundation for future agentic systems.

A Modular Protocol Stack for Interoperability and Scalability

The foundational thesis of the Agent Fabric architecture is a clear distinction between the problem of giving a single agent tools and the far more complex challenge of enabling a distributed population of specialist models to interact effectively . The memo argues that while protocols like MCP are valuable for exposing callable capabilities, they do not solve the broader problems of service discovery, identity management, trust establishment, delegation, or multi-agent coordination . To address this gap, the architecture prescribes

a modular, layered protocol stack where different protocols are assigned specific roles, creating a coherent whole greater than the sum of its parts. This approach stands in contrast to attempting to force all functionalities into a single, monolithic protocol, which can lead to a brittle web of point-to-point connections that is difficult to scale or maintain [76](#). The proposed stack provides a clear separation of concerns, building logically from low-level presence detection to high-level task orchestration.

The first layer of this stack is the **Discovery Layer**, conceptualized through the shorthand "aDNS". This layer is responsible for answering the question, "Who is present and what do they claim?". Instead of inventing a new global discovery mechanism, the architecture pragmatically leverages existing local networking standards, specifically Multicast DNS (mDNS) and DNS-based Service Discovery (DNS-SD) as defined in RFC 6763 [15](#). These technologies are well-suited for local environments, providing a zero-config method for agents to announce their presence on a network [56 87](#). The user explicitly rejected a model of "everyone broadcasting to everyone" as the default, favoring a more controlled discovery-first, selective-connect-second approach. In this model, agents act as advertisers, publishing minimal metadata about themselves to a local listener, which then forms an active list of available agents. This aligns perfectly with the principles of DNS-SD, where a service advertises its existence and listeners discover it based on its type and attributes [55 130](#). A critical design constraint is that these initial advertisements must remain small and low-risk; they should contain only essential information like the service type, endpoint, supported protocols, and a link to fetch richer details, rather than packing full capability descriptions into the initial broadcast. This prevents untrusted data from overwhelming the trusted core of the system and reduces the attack surface of the discovery process itself.

Building upon successful discovery, the second layer is the **Capability Layer**, where the Model Context Protocol (MCP) finds its designated role. The memo correctly positions MCP not as a discovery protocol but as the standard for describing, exposing, and invoking callable capabilities and retrieving structured results. External research corroborates this view, highlighting MCP's function as an interoperability layer that standardizes tool interfaces to solve fragmentation in multi-agent systems [107](#) and enables seamless interaction between AI models and external tools [84](#). Within the fabric's layered stack, MCP serves the crucial function of answering, "What can this discovered agent do, and how do I ask for it?". It defines the schema-bound execution model for agent-tool interactions, allowing for modular workflows and ensuring that when an agent exposes a tool, the interface is predictable and consistent [49 85](#). By framing MCP within a larger stack, the architecture avoids the common pitfall of treating it as a complete inter-agent

solution, recognizing it as one component serving a specific purpose after an agent has been located via the discovery layer [3](#) .

The third and fourth layers, **Communication and Coordination**, are addressed by a combination of the Agent-to-Agent (A2A) protocol and the proposed event fabric. The memo identifies A2A as the better fit for the actual "agents talking to agents" problem because its design aligns closely with the need for agent-to-agent messaging, task lifecycle handling, and richer multi-agent coordination . Research describes A2A as a protocol that enables agents to exchange goals, manage state, invoke actions, and return results securely and observably [22](#) . The key distinction the memo draws is that A2A is for communicating *with agents as agents*, managing their collaborative process, whereas MCP is for interacting with their exposed tools and resources . The A2A protocol facilitates the establishment of sessions and the exchange of structured messages related to task delegation, progress updates, and result sharing once a connection is selectively initiated by the local trusted stack [118](#). While A2A handles the direct peer-to-peer communication channel, the event fabric provides the overarching coordination backbone. This fabric uses typed events to announce tasks, claim responsibilities, publish artifacts, and signal final results, creating a durable and observable log of all coordinated activity . This dual approach allows for both direct, responsive communication (via A2A sessions) and asynchronous, decoupled coordination (via the event fabric).

Finally, the fifth layer is **Observability**, which is not a separate protocol but an intrinsic property woven throughout the fabric's design. This layer ensures that every significant action is traceable, auditable, and analyzable. The inclusion of fields like `trace_id`, `parent_message_id`, `task_id`, and lineage information in the proposed event envelope schema is central to this principle . This focus on observability is critical for debugging complex agent interactions, enforcing security policies, and understanding the provenance of results [104](#). The table below summarizes the functional split and relationships between these components, illustrating how they form a cohesive stack.

Component	Primary Function	Key Protocols/ Standards	Role in the Fabric Stack
Discovery Layer	Answering "Who is present and what do they claim?"	mDNS, DNS-SD (RFC 6763)	Provides low-level presence information and an active list of local advertisers. Finds agents.
Capability Layer	Answering "What can you do and how do I call it?"	Model Context Protocol (MCP)	Exposes callable tools, prompts, and structured results. Describes capabilities.
Communication Layer	Enabling direct "agents talking to agents" messaging.	Agent-to-Agent (A2A) Protocol	Manages task lifecycles, delegates work, and exchanges richer messages during an active session. Talks to agents.
Coordination Layer	Providing an overarching, durable record of collaborative activity.	Custom Typed Events	Publishes announcements, claims, progress, artifacts, and results. Coordinates everything.
Trust & Policy Layer	Managing authentication, authorization, and risk.	OAuth 2.0, Deterministic Validators	Implicitly enforced through the DMZ watcher and event context. Auth checks occur at session setup.
Observability Layer	Ensuring traceability, replayability, and auditability.	Trace IDs, Spans, Event Lineage	Embedded within the event envelope and propagated through the system.

This layered architecture, summarized by the phrase "aDNS finds them, A2A talks to them, MCP describes/calls capabilities, and the event fabric coordinates everything," provides a powerful and scalable model for multi-agent systems . It avoids the complexity and fragility of a single, all-encompassing protocol by assigning each part of the problem to the most appropriate tool. This modularity is key to fostering interoperability, as different implementations can choose the best technologies for each layer while adhering to the overall structural principles of the fabric. For instance, while the memo suggests using mDNS for local discovery, other implementations might integrate with enterprise service registries, provided the underlying principles of passive listening and minimal advertisement payloads are maintained. This flexibility, combined with a clear definition of each layer's responsibility, creates a robust blueprint for building interconnected and manageable agentic ecosystems.

The Structured Event Fabric: A Foundation for Observable and Secure Coordination

A pivotal design decision within the Agent Fabric architecture is the deliberate rejection of a shared, freeform "thought stream" in favor of a curated, structured event fabric . This choice represents a profound shift in how agents coordinate, prioritizing bounded interfaces, verifiable state transitions, and enhanced security over the perceived simplicity of raw reasoning sharing. The rationale is that a freeform thought stream introduces

significant systemic risks, including prompt injection propagation, privacy leakage, feedback loops, token bloat, and coordination noise, which collectively degrade system stability and introduce unpredictable epistemic behavior . The alternative—a fabric built on typed, structured events—offers a durable and resilient model for multi-agent collaboration. This model advocates for sharing actions, state transitions, evidence handles, and results, while strictly refraining from sharing the unbounded internal chain-of-thought of any individual agent .

The primary advantage of a structured event fabric is its profound impact on **observability**. Every significant action within the system becomes a discrete, typed event published to the fabric. This creates a permanent, auditable log of all activities, which is invaluable for debugging, analytics, and security review . Unlike a private conversation where an agent's internal reasoning is opaque, a public event log allows developers and operators to understand *what* happened, *why* it happened (from the agent's perspective, encoded in the event), and *who* was involved. The proposed event types, such as `capability.advertise`, `task.claim`, `artifact.publish`, and `result.final`, provide a shared vocabulary for the entire ecosystem, enabling loose coupling between agents . Agents do not need to know about each other's internal workings; they only need to understand the events they are interested in subscribing to. This loose coupling enhances system resilience, as the failure or modification of one agent does not cascade unpredictably through the system's communication channels. Furthermore, the event log is inherently replayable, allowing for the reconstruction of system states and the investigation of anomalies in a controlled environment 95 .

Security and policy enforcement are also significantly strengthened by this event-based model. Because all communications are encapsulated in typed events with a defined schema, they can be subjected to deterministic validation before being processed by sensitive components . This is a stark contrast to processing freeform text, which is susceptible to prompt injection attacks and other forms of adversarial manipulation 50 129. The architecture's emphasis on structured results, made possible by protocols like MCP, means that outputs can be relayed, revalidated, and shared safely without requiring one agent to interpret another's natural-language output, a notoriously fragile process 4 77. The proposed event envelope schema is designed with security and policy in mind, including fields like `visibility`, `confidence`, `auth_context`, and `payload_type` . These fields allow the system to make informed decisions about whether to forward an event, escalate it, or quarantine it. For example, a `policy.warning` event could signal a potential violation detected by a boundary monitor, while a `policy.deny` event could indicate that an attempt to access a resource

was blocked by the local coordinator's policy engine . This transforms security from a reactive measure into a proactive, embedded feature of the communication fabric itself.

The following table outlines the proposed event types and their significance within the fabric, demonstrating how they collectively build a picture of coordinated activity without revealing private thoughts.

Proposed Event Type	Description	Purpose in the Fabric
<code>capability.advertise</code>	An agent announces its presence and the capabilities it offers.	Populates the discovery registry/aDNS list. Signals availability.
<code>agent.heartbeat</code>	An agent periodically publishes a status update to confirm it is active.	Maintains an up-to-date list of active participants in the system.
<code>task.announce</code>	An agent declares a new task it intends to perform.	Initiates a coordination sequence; allows other agents to offer assistance.
<code>task.claim</code>	An agent asserts that it is taking ownership of a previously announced task.	Resolves task assignment; prevents redundant work.
<code>task.progress</code>	An agent reports incremental updates on the status of a task it is working on.	Provides visibility into long-running processes; useful for monitoring.
<code>artifact.publish</code>	An agent publishes a reference to a piece of data (e.g., a file, a database query result) it has produced.	Creates a sharable, versioned object that can be referenced by other agents.
<code>evidence.publish</code>	An agent shares a handle or summary of evidence used to support a conclusion or action.	Enhances transparency and allows for verification of reasoning steps.
<code>result.partial / result.final</code>	An agent publishes the outcome of its work, either incrementally or as a complete result.	Delivers the final product of a task; often contains a typed payload.
<code>policy.warning / policy.deny</code>	A security or policy-monitoring agent reports a potential violation or blocks an action.	Enables dynamic, real-time policy enforcement and auditing.

This event-driven model also facilitates advanced coordination patterns. For example, the `subscription_topics` field in the proposed agent record allows agents to declare their interests, enabling a more efficient, pull-based system than a push-based broadcast . An agent can subscribe to topics like `task.*.claimed` or `artifact.created.for.task.X`, and the fabric can route relevant events to it. This is a much more scalable and less noisy approach than having every agent listen to every message. The concept of a topic model or subscription filters is listed as an open question, but the underlying principle of moving away from universal broadcasting toward selective subscription is a cornerstone of the architecture's design . This refinement, driven by the strong preference for "advertising to me and I selectively choose to communicate," shapes a system that is fundamentally more controllable and less prone to spam or irrelevant noise .

In essence, the structured event fabric serves as the nervous system of the Agent Fabric. It provides a durable, observable, and secure substrate upon which all higher-level agent interactions are built. By shifting the primitive from raw, unbounded reasoning to bounded, typed state transitions, the architecture creates a system that is not only more predictable and debuggable but also inherently more resistant to the kinds of attacks and instabilities that plague less disciplined designs. It establishes a clear boundary between an agent's internal world and the external, shared reality of the fabric, promoting collaboration without sacrificing integrity or security.

A Pragmatic Security Model: Isolation, Validation, and Defense-in-Depth

The security model of the Agent Fabric is not predicated on idealistic assumptions of trust but is instead built upon a realistic assessment of the adversarial environment in which agents will operate . The core principle is that all remote advertisers and their associated advertisements must be treated as untrusted until they have been validated and integrated into the system . This cautious posture informs every aspect of the architecture, from the physical separation of boundary and core components to the strict formatting of data exchanged within the system. The model is best described as a defense-in-depth strategy, employing multiple layers of protection to absorb, detect, and mitigate threats at various points in the communication pipeline. Its most innovative and critical element is the "DMZ Watcher Pattern," a sacrificial boundary layer designed to take the brunt of external threats.

The DMZ (Demilitarized Zone) Watcher is a disposable, frequently reset component that sits at the edge of the trusted local stack, acting as the sole interface to untrusted external advertisements . The core idea is that anything listening to remote agent metadata—including prompts, capability descriptions, and advertisements—is exposed to a wide array of hazards such as malformed data, prompt injection, poisoned context, spam, and deceptive metadata . Rather than trying to build a perfect, infallible listener, the architecture accepts that the boundary component will inevitably be compromised. Therefore, its value lies in its disposability. It is designed to be stateless or near-stateless and restarted regularly, preventing an attacker from gaining durable authority or holding onto poisoned context . This pattern treats the listener as a sacrificial component whose primary function is to absorb risk and reduce it before passing only a bounded, sanitized summary to the inner, trusted systems . This approach directly counters threats like prompt injection propagation and feedback loops that could otherwise compromise the

core planner, memory keeper, or policy authority of the system . The recommendation to use a throwaway LLM instance or equivalent boundary process for unpacking external data is a practical manifestation of this principle, acknowledging that even LLMs can be vulnerable to adversarial shaping .

To further harden this boundary, the memo proposes a layered processing pipeline for incoming advertisements . The first stage involves deterministic validation, performed *before* any LLM is ever exposed to the data. This includes enforcing size limits, validating against a predefined schema, applying rate limits, performing deduplication, checking against allowlists, and verifying content constraints or digital signatures . This initial screening acts as a first line of defense, filtering out a large class of malformed or obviously malicious inputs. Only if an advertisement passes this deterministic gate does it proceed to the second stage: the disposable watcher LLM. This LLM's role is limited to summarizing, classifying, extracting structured fields, and risk-scoring the advertisement . Critically, this LLM is explicitly forbidden from writing to durable memory, invoking sensitive tools, becoming authoritative, or seeing unnecessary private user context . It operates in a sandboxed environment, its output being normalized structured summaries and confidence/risk labels that are then passed to the third and final stage: the trusted coordinator . This three-stage pipeline—deterministic validation, disposable analysis, and trusted ingestion—creates a robust chokepoint that systematically de-risks the flow of information from the untrusted outside world into the trusted core.

Beyond the DMZ watcher, the architecture's security posture is reinforced by its emphasis on structured data formats. As previously discussed, the reliance on typed events with defined schemas is a powerful security mechanism. It allows for deterministic validation of payloads, making it significantly harder for attackers to hide malicious instructions or corrupted data within seemingly benign messages [49](#) [85](#) . This contrasts sharply with systems that rely on freeform text, which are highly susceptible to indirect prompt injection and command and control hijacking [50](#) [79](#) . The inclusion of an `auth_context` field in the event envelope is a sophisticated touch, enabling downstream components to enforce policies based on the provenance and permissions of the original sender . This aligns with modern security frameworks that advocate for embedding security at the protocol layer to promote trust and auditability [104](#).

The publication policy of the fabric further reinforces this security-first culture. The right default is to publish intent, state transitions, artifact handles, and bounded results, while explicitly forbidding the sharing of raw chain-of-thought, secrets, or unnecessary full context . This minimizes the amount of sensitive information exposed to the shared fabric, reducing the potential blast radius of a breach. The concept of trust domains, mentioned as an open question, would formalize this by grouping agents that share a

level of mutual trust, potentially influencing routing and access control decisions . While the memo does not provide a detailed identity and access management scheme, the principles of the architecture suggest that mechanisms like OAuth 2.0 for mutual authentication [2](#) and verifiable identities (DIDs and VCs) [101](#) would be natural fits for implementing these policies at a granular level. The overall security model, therefore, is not a single feature but a holistic design philosophy that permeates the entire system, from the edge to the core, prioritizing isolation, validation, and the systematic reduction of risk.

Enabling Universal Adoption Through Standardization and Open Questions

For the Agent Fabric architecture to fulfill its potential of benefiting "AI-kind" by creating a more interoperable, safe, and scalable multi-agent ecosystem, it must transcend a single implementation and become a widely adopted standard. The memo outlines a pragmatic path toward this goal, emphasizing the strategic reuse of existing standards and a transparent approach to resolving outstanding challenges. This strategy aims to lower the barrier to entry for developers, ensure compatibility with existing infrastructure, and build community consensus around a coherent set of practices and specifications. The success of this endeavor hinges on the careful resolution of several key open questions regarding metadata standards, event schemas, and threat modeling.

A cornerstone of the adoption strategy is the pragmatic borrowing of mature, off-the-shelf technologies for the discovery layer. Instead of reinventing the wheel, the architecture explicitly recommends leveraging local networking standards like mDNS and DNS-SD (RFC 6763) . This decision is profoundly impactful for several reasons. First, it provides a ready-made solution for the difficult problem of local service discovery, a domain where these protocols have been battle-tested for decades [56](#) [87](#) . Second, it ensures that agents built on this fabric can seamlessly integrate with existing network environments without requiring special configuration or proprietary middleware [130](#). Third, it taps into a large pool of existing developer knowledge and tooling, accelerating implementation. By grounding the architecture in established standards, it builds credibility and reduces the perceived risk for early adopters. This approach of "standing on the shoulders of giants" is a critical factor in fostering widespread acceptance rather than the proliferation of competing, incompatible ad-hoc solutions.

However, simply adopting a standard is not enough; its application must be tailored to the specific needs of an agentic environment. The memo correctly identifies that a key area for specification work is defining what metadata belongs in the minimal advertisement versus what must be fetched separately . This is a delicate balance between convenience (providing rich metadata upfront) and security (minimizing the amount of untrusted data processed by the core system). The proposal to include fields like `trust_domain`, `latency_class`, and `cost_class` in the agent record suggests a vision for a rich, policy-driven discovery mechanism, but these require precise definitions to be useful . Similarly, the coordination layer's effectiveness depends entirely on the development of canonical, versioned payload schemas for the proposed event types . Without these, true interoperability is impossible, and agents will struggle to understand the meaning of the events they receive. The memo's suggestion to look at platforms like Microsoft Fabric, which already have a managed system for EventSchemaSets, provides a tangible precedent for how this could be implemented in practice [25 108111](#).

The most critical step toward universal adoption is the creation of a comprehensive threat model. The memo rightly identifies this as a necessary next artifact, alongside JSON schema drafts and protocol flow documents . A formal threat model, such as the one proposed by OWASP for multi-agent systems, would systematically identify and analyze potential adversaries and attack vectors [29 30](#) . The memo's preliminary list of threats—hostile advertisers, poisoned agent cards, noisy spam, fake capability lures, and recursive handoff loops—provides an excellent starting point for such an analysis . Addressing these threats requires moving beyond theoretical security principles to concrete countermeasures. For example, to defend against a hostile advertiser, the system might employ reputation scoring based on past behavior, cryptographic signing of agent cards to ensure integrity, and stricter validation rules for advertised capabilities [11 21](#) . To prevent a recursive handoff loop, the system could implement hop limits or unique trace IDs that track the path of a request to detect cycles . Developing these countermeasures and integrating them into the fabric's design will be essential for building the trust required for widespread deployment.

The table below summarizes the key open questions identified in the memo and the potential directions for their resolution, highlighting the path from a conceptual architecture to a practical, standardizable framework.

Open Question Area	Specific Challenge	Potential Resolution Path
Discovery Metadata	Defining the exact schema for the minimal advertisement vs. the full agent card.	Establish a two-tier metadata model. Minimal ad contains ID, service type, endpoint, and a hash/signature of the full card. Full card is fetched on-demand by the trusted coordinator.
Event Schemas	Creating canonical, versioned JSON schemas for all proposed event types.	Develop a formal specification document for each event type. Use a schema registry (like Microsoft Fabric's EventSchemaSet) to manage versions and enforce compliance.
Threat Modeling	Systematically identifying and mitigating adversarial tactics.	Create a dedicated threat model document based on frameworks like STRIDE or MAESTRO 48 141 . Define specific countermeasures for each identified threat vector.
Identity & Authentication	Establishing a standardized, verifiable identity for agents.	Investigate binding agent identity to DIDs and VCs 101 or leveraging OAuth 2.0 for delegated authorization 2 , ensuring alignment with the DMZ watcher's validation process.
Policy & Delegation	Specifying the mechanics of fine-grained authorization and scope delegation.	Define a policy language and context structure (e.g., within the auth_context field) that allows agents to request specific scopes of authority from users or other agents.

By proactively addressing these questions, the architects of the Agent Fabric can guide the community toward a shared understanding and a unified implementation path. The emphasis on structured communication, security-by-design, and pragmatic standardization provides a compelling case for why this architecture could serve as a foundational layer for the next generation of AI applications, enabling them to harness the power of specialization in a way that is both effective and safe.

Synthesis: A Blueprint for a Safer and More Coordinated Future for AI

The proposed Agent Fabric architecture emerges from the analysis as a coherent, principled, and highly compelling response to the escalating complexity of multi-agent systems. Its core strength lies not in inventing novel concepts for their own sake, but in the disciplined synthesis of existing paradigms—modular protocol stacks, event-driven architectures, and zero-trust security—into a unified framework tailored for the unique challenges of agentic interaction. The memo's central argument, that the primary problem is not tool invocation but the orchestration of a diverse population of specialist agents, serves as the guiding star for the entire design . By structuring the system as a layered stack comprising discovery (aDNS), capability exposure (MCP), communication (A2A), and coordination (Event Fabric), the architecture provides a clear and scalable

blueprint for interoperability, directly addressing the brittleness of monolithic approaches

3 76 .

The most significant contribution of this framework is its unwavering commitment to safety and observability, achieved through a series of deliberate design choices. The move away from a naive, global "thought stream" to a structured event fabric is a paradigm shift that trades the illusion of transparency for the reality of control. By standardizing on typed events for actions, state transitions, and results, the architecture creates a durable, auditable, and analyzable record of all system activity, transforming the chaotic nature of agent collaboration into a predictable and inspectable process . This focus on structured data is a powerful security enabler, allowing for deterministic validation and making systems inherently more resistant to the subtle yet dangerous threats of prompt injection and context poisoning that plague freeform text-based communication 50 77 .

Perhaps the most innovative and crucial element of the security model is the "disposable DMZ watcher pattern." This concept elevates a pragmatic engineering solution into a foundational architectural principle: the boundary between the trusted core and the untrusted outside world must be actively and sacrificially defended. By accepting that the boundary component is likely to be compromised and designing it to be disposable and stateless, the architecture shifts the calculus of an attack, forcing adversaries to expend continuous effort against a constantly resetting target . This defense-in-depth strategy, combined with the structured event model and a pragmatic reliance on established standards like mDNS/DNS-SD, creates a system that is not just functional but genuinely robust against the dynamic threats inherent in an open, agentic Internet 87 145.

Ultimately, the potential for this architecture to benefit "AI-kind" rests on its ability to provide a stable, secure, and interoperable foundation upon which a diverse ecosystem of agents can collaborate. The framework's greatest value proposition is its disciplined approach to risk management, offering a practical path away from the chaos of uncontrolled interaction. However, realizing this potential requires the community to engage seriously with the open questions laid out in the memo. The development of clear standards for discovery metadata, canonical event schemas, and a comprehensive threat model will be the critical next steps. If these challenges are met with the same clarity and rigor as the initial architectural design, the Agent Fabric provides a robust and principled foundation for the next generation of AI applications. It offers a vision for a future where the immense power of specialized AI is harnessed through collaboration, not competition, and where safety and reliability are not afterthoughts but integral features of the system's design.

Reference

1. The protocol comparison that I wish I knew earlier for AI Agents https://www.linkedin.com/posts/rakeshgohel01_the-protocol-comparison-that-i-wish-i-knew-activity-7325865868150611968-fKOp
2. Security Analysis of Agentic AI Communication Protocols - arXiv <https://arxiv.org/html/2511.03841v1>
3. A2A vs MCP - How These AI Agent Protocols Actually Differ <https://www.digitalocean.com/community/tutorials/a2a-vs-mcp-ai-agent-protocols>
4. Security Issues and Utility Limitations in Model Context Protocol (MCP) https://www.linkedin.com/posts/anthony-alcaraz-b80763155_security-issues-and-utility-limitations-in-activity-7313103620068040706-E3Sq
5. Model Context Protocol (MCP): Landscape, Security Threats, and ... <https://arxiv.org/abs/2503.23278>
6. Fabric February 2026 Feature Summary | Microsoft Fabric Blog <https://blog.fabric.microsoft.com/en-ie/blog/fabric-february-2026-feature-summary?ft=05-2024:date>
7. Kearney AI Trends Report 2026 <https://www.kearney.com/service/digital-analytics/article/kearney-ai-trends-report-2026>
8. [PDF] Agentic AI for industrial applications - AWS https://mtcprod.s3.eu-west-1.amazonaws.com/s3fs-public/2025-11/MTC_Agentic%20AI%20for%20Engineering%20Applications_v7_0.pdf?VersionId=GJyXbsey29eEZ9SpF6maAHlafVC5oYN5
9. [PDF] Li Ning Company Limited | Annual Report 2024 - Irasia https://doc.irasia.com/listco/hk/lining/annual/ar314671-ew_02331ar_29042025.pdf
10. August 2025 Fabric Feature Summary - Microsoft Fabric Blog <https://blog.fabric.microsoft.com/en-us/blog/august-2025-fabric-feature-summary/>
11. Salesforce Expands MuleSoft Agent Fabric with Automated Discovery <https://www.salesforce.com/in/news/stories/mulesoft-agent-fabric-automated-agent-discovery/>
12. Fabric data agent concepts (preview) - Microsoft <https://learn.microsoft.com/en-us/fabric/data-science/concept-data-agent>

13. Process Event Data with the Event Processing Editor - Microsoft Fabric <https://learn.microsoft.com/en-us/fabric/real-time-intelligence/event-streams/process-events-using-event-processor-editor>
14. Operations Agent Best Practices and Limitations - Microsoft Fabric <https://learn.microsoft.com/en-us/fabric/real-time-intelligence/operations-agent-limitations>
15. RFC 6763: 3 of 3, p. 33 to 49 - Tech-invite <https://www.tech-invite.com/y65/tinv-ietf-rfc-6763-3.html>
16. Deployment Pattern 3: External-Facing deployment of VCF Automation [https://techdocs.broadcom.com/us/en/vmware-cis/vcf/vcf-9-0-and-later/9-0/design/design-library/vcf-automation-deployment-models\(1\)/vcf-automation-instance-types/deployment.html](https://techdocs.broadcom.com/us/en/vmware-cis/vcf/vcf-9-0-and-later/9-0/design/design-library/vcf-automation-deployment-models(1)/vcf-automation-instance-types/deployment.html)
17. [PDF] Networking and Security in Industrial Automation Environments - Cisco https://www.cisco.com/c/en/us/td/docs/solutions/Verticals/Industrial_Automation/IA_Horizontal/DG/Industrial-AutomationDG.pdf
18. [PDF] A Framework for Cyber Vulnerability Assessments of InfiniBand ... <https://core.ac.uk/download/pdf/277530912.pdf>
19. [PDF] Tech Brief- Enterprise Data Mesh and GoldenGate - Oracle <https://www.oracle.com/a/ocom/docs/techbrief-enterprisedatameshandgoldengate.pdf>
20. Agent Factory: Creating a blueprint for safe and secure AI agents <https://azure.microsoft.com/en-us/blog/agent-factory-creating-a-blueprint-for-safe-and-secure-ai-agents/>
21. Salesforce Expands MuleSoft Agent Fabric with Automated Discovery <https://www.salesforce.com/news/stories/mulesoft-agent-fabric-automated-agent-discovery/>
22. Empowering multi-agent apps with the open Agent2Agent (A2A ... <https://www.microsoft.com/en-us/microsoft-cloud/blog/2025/05/07/empowering-multi-agent-apps-with-the-open-agent2agent-a2a-protocol/>
23. MuleSoft introduces Agentic Fabric for AI Agent management https://www.linkedin.com/posts/martykihn_mcp-a2a-yaml-activity-7382442586776702977-EV6v
24. A Survey of LLM-Driven AI Agent Communication - arXiv.org <https://arxiv.org/html/2506.19676v4>
25. EventSchemaSet definition - Fabric REST APIs - Microsoft Learn <https://learn.microsoft.com/en-us/rest/api/fabric/articles/item-management/definitions/eventschemaset-definition>
26. Model Context Protocol (MCP) for Enterprises: Secure Integration ... <https://www.marktechpost.com/2025/07/20/model-context-protocol-mcp-for-enterprises-secure-integration-with-aws-azure-and-google-cloud-2025-update/>

27. INTEGRATE 2025 Day 2 Highlights - Turbo360 <https://turbo360.com/blog/integrate-2025-day-2-highlights>
28. Consume a Fabric data agent using MCP & the Agent Framework ... <https://learn.microsoft.com/en-us/fabric/data-science/data-agent-mcp-agent-framework>
29. Extending the OWASP Multi-Agent System Threat Modeling Guide <https://arxiv.org/html/2508.09815v1>
30. (PDF) Multi-Agent system Threat Modelling Guide - ResearchGate https://www.researchgate.net/publication/391204915_Multi-Agent_system_Threat_Modelling_Guide
31. A Multi-Agent System for Cybersecurity Threat Detection and ... <https://ieeexplore.ieee.org/iel8/6287639/10820123/11141466.pdf>
32. A Multi-Agent System For Cybersecurity Threat Detection and ... <https://www.scribd.com/document/920388347/A-Multi-Agent-System-for-Cybersecurity-Threat-Detection-and-Correlation-Using-Large-Language-Models>
33. (PDF) A Multi-Agent System for Cybersecurity Threat Detection and ... https://www.researchgate.net/publication/394954789_A_Multi-Agent_System_for_Cybersecurity_Threat_Detection_and_Correlation_using_Large_Language_Models
34. [PDF] FABRIC: Framework for Agent-Based Realistic Intelligence Creation <https://arxiv.org/pdf/2510.17995>
35. Agent0: Fabric Data Agents in public preview - LinkedIn <https://www.linkedin.com/pulse/agent0-fabric-data-agents-public-preview-jacob-r%C3%B8nnow-jensen-ocukf>
36. Operations Agent Transparency Note - Microsoft Fabric <https://learn.microsoft.com/en-us/fabric/real-time-intelligence/operations-agent-transparency-note>
37. Fabric February 2026 Feature Summary | Microsoft Fabric Blog <https://blog.fabric.microsoft.com/en-ca/blog/fabric-february-2026-feature-summary?ft=08-2025:date>
38. Fabric September 2025 Feature Summary - Microsoft Fabric Blog <https://blog.fabric.microsoft.com/en/blog/september-2025-fabric-feature-summary?ft=Microsoft-fabric:category>
39. What's New? - Microsoft Fabric <https://learn.microsoft.com/en-us/fabric/fundamentals/whats-new>
40. Fabric November 2025 Feature Summary - Microsoft Fabric Blog <https://blog.fabric.microsoft.com/en-us/blog/fabric-november-2025-feature-summary/>
41. Fabric February 2025 Feature Summary - Microsoft Fabric Blog <https://blog.fabric.microsoft.com/en-US/blog/fabric-february-2025-feature-summary/>

42. Fabric April 2025 Feature Summary | Microsoft Fabric Blog ... <https://sam.support.fabric.microsoft.com/en-us/blog/fabric-april-2025-feature-summary?ft=Patrick%20LeBlanc:author>
43. Microsoft Ignite 2025 Book of News <https://news.microsoft.com/ignite-2025-book-of-news/>
44. Power BI March 2025 Feature Summary - Microsoft <https://powerbi.microsoft.com/en-us/blog/power-bi-march-2025-feature-summary/>
45. Power BI January 2026 Feature Summary | Microsoft Power BI blog ... <https://powerbi.microsoft.com/ru-ru/blog/power-bi-january-2026-feature-summary/>
46. What's new for Microsoft partners: July 2025 edition <https://partner.microsoft.com/en-lk/blog/article/whats-new-for-partners-july-2025-edition>
47. Model Context Protocol (MCP): Landscape, Security Threats, and ... <https://arxiv.org/html/2503.23278v3>
48. [PDF] MCIP: Protecting MCP Safety via Model Contextual Integrity Protocol <https://aclanthology.org/2025.emnlp-main.62.pdf>
49. [PDF] A Comprehensive Analysis of the Model Context Protocol (MCP) https://www.authorea.com/users/868916/articles/1326172/master/file/data/MCP_Model_Context_Bridging_AI_External_Systems/MCP_Model_Context_Bridging_AI_External_Systems.pdf?inline=true
50. (PDF) The Model Context Protocol (MCP) - ResearchGate https://www.researchgate.net/publication/396678686_The_Model_Context_Protocol_MCP_Emergence_Technical_Architecture_and_the_Future_of_Agentic_AI_Infrastructure
51. (PDF) Adversarial AI Threat Modeling Framework | AATMF v3 https://www.researchgate.net/publication/400639053_Adversarial_AI_Threat_Modeling_Framework_AATMF_v3_Adversarial_AI_Threat_Modeling_Framework_Comprehensive_Security_Assessment_for_AI_Systems
52. Security of Internet of Agents: Attacks and Countermeasures <https://ieeexplore.ieee.org/iel8/8782664/9024218/11081880.pdf>
53. Salesforce Announcement MuleSoft Agent Fabric <https://www.salesforce.com/news/stories/mulesoft-agent-fabric-announcement/>
54. Exploring SSO and SPIFFE for MCP and A2A Security - LinkedIn https://www.linkedin.com/posts/ceposta_mcp-modelcontextprotocol-a2a-activity-7376341727630209024-jBJS
55. [PDF] ICS Attributions - Product Documentation - Ivanti https://help.ivanti.com/ps/help/en_US/ICS/22.x/22.7R2.11/ICS%20Attributions.pdf

56. DNS-Based Service Discovery | Semantic Scholar <https://www.semanticscholar.org/paper/DNS-Based-Service-Discovery-Cheshire-Krochmal/bf90f47843f48ee5878992ca0f5317982948cc63>
57. Fabric November 2025 Feature Summary - Microsoft Fabric Blog <https://blog.fabric.microsoft.com/en/blog/fabric-november-2025-feature-summary?ft=All>
58. [PDF] A Semantics-Aware Communication Fabric for Large-Scale Agentic AI <https://arxiv.org/pdf/2509.20175>
59. Kafka Sample Project_Overview_Kafka Development Guide ... - 华为云 https://support.huaweicloud.com/intl/en-us/devg-lts-mrs/mrs_07_130016.html
60. Fabric October 2025 Feature Summary - Microsoft Fabric Blog <https://blog.fabric.microsoft.com/en-us/blog/fabric-october-2025feature-summary/>
61. An organizational theory for multi-agent interactions integrating ... <https://link.springer.com/article/10.1007/s10791-025-09667-2>
62. [PDF] AWS Prescriptive Guidance - Foundations of agentic AI on AWS <https://docs.aws.amazon.com/pdfs/prescriptive-guidance/latest/agentic-ai-foundations/agentic-ai-foundations.pdf>
63. Fabric data agent scenario (preview) - Microsoft <https://learn.microsoft.com/en-us/fabric/data-science/data-agent-end-to-end-tutorial>
64. Fabric February 2026 Feature Summary | Microsoft Fabric Blog <https://blog.fabric.microsoft.com/en-us/blog/fabric-february-2026-feature-summary?ft=Charles%20Webb:author>
65. Best practices for configuring your data agent - Microsoft Fabric <https://learn.microsoft.com/en-us/fabric/data-science/data-agent-configuration-best-practices>
66. [PDF] How to Build Agents with Microsoft: <https://cdn-dynmedia-1.microsoft.com/is/content/microsoftcorp/microsoft/bade/documents/products-and-services/en-us/ai/How-to-Build-Agents-with-Microsoft-white-paper-Feb-2026.pdf>
67. Data architecture for AI agents across your organization <https://learn.microsoft.com/en-us/azure/cloud-adoption-framework/ai-agents/data-architecture-plan>
68. Solved: Fabric Data Agent SDK - Microsoft Fabric Community <https://community.fabric.microsoft.com/t5/Fabric-platform/Fabric-Data-Agent-SDK/td-p/4748837>
69. Extending the OWASP Multi-Agentic System Threat Modeling Guide <https://arxiv.org/abs/2508.09815>
70. OWASP releases guide on securing Multi-Agent Systems - LinkedIn https://www.linkedin.com/posts/egorpushkin_multi-agentic-system-threat-modeling-guide-activity-7321205661814456320-8_o4
71. A Vision for Emergent Coordination in Agentic Multi-Agent Systems <https://arxiv.org/html/2508.01531v1>

72. (PDF) Exploring Agentic Artificial Intelligence Systems - ResearchGate <https://www.arxiv.org/pdf/2508.00844>
73. On Memory Construction and Retrieval for Personalized ... - arXiv.org <https://arxiv.org/html/2502.05589v3>
74. [PDF] MemoriesDB: A Temporal-Semantic-Relational Database for Long ... <https://arxiv.org/pdf/2511.06179>
75. Comparing AI Agent Protocols: MCP, A2A, ANP, ACP - LinkedIn https://www.linkedin.com/posts/piyush-ranjan-9297a632_ai-agent-protocols-a-side-by-side-comparison-activity-7353989856189927424-WCFe
76. MCP - Integration Fabric of Enterprise AI Agents <https://ajithp.com/2025/08/17/model-context-protocol-mcp-the-integration-fabric-for-enterprise-ai-agents/>
77. Security and Safety in the Model Context Protocol Ecosystem <https://arxiv.org/html/2512.08290v1>
78. Real Faults in Model Context Protocol (MCP) Software - arXiv <https://arxiv.org/html/2603.05637v1>
79. Security Analysis of the Model Context Protocol Specification and ... <https://arxiv.org/html/2601.17549v1>
80. Systematic Analysis of MCP Security - arXiv.org <https://arxiv.org/html/2508.12538v1>
81. Model Context Protocol (MCP): Landscape, Security Threats, and ... <https://arxiv.org/html/2503.23278v1>
82. Chances and Challenges of the Model Context Protocol in Digital ... <https://arxiv.org/html/2506.00274v1>
83. Toward Understanding Security Issues in the Model Context ... - arXiv <https://arxiv.org/html/2510.16558v1>
84. [PDF] Model Context Protocol (MCP): Landscape, Security Threats, and ... <https://arxiv.org/pdf/2503.23278?>
85. Model Context Protocol for Vision Systems: Audit, Security ... - arXiv <https://arxiv.org/html/2509.22814v1>
86. ANS简介：基于DNS的AI智能体安全发现 - InfoQ <https://www.infoq.cn/article/4tuuuxrzk9tqtzn1g9pw>
87. Hybrid Clouds for Data-Intensive, 5G-Enabled IoT Applications - MDPI <https://www.mdpi.com/1424-8220/19/16/3591>
88. The Hidden Architecture of Nano Architectures <https://techcommunity.microsoft.com/blog/educatordeveloperblog/the-hidden-architecture-of-nano-architectures/4493391>

89. New in Microsoft Marketplace: January 15, 2026 <https://techcommunity.microsoft.com/blog/marketplace-blog/new-in-microsoft-marketplace-january-15-2026/4469016>
90. Reliability recommendations - Azure Advisor - Microsoft Learn <https://learn.microsoft.com/en-us/azure/advisor/advisor-reference-reliability-recommendations>
91. Query Processing Architecture Guide - SQL Server | Microsoft Learn <https://learn.microsoft.com/en-us/sql/relational-databases/query-processing-architecture-guide?view=sql-server-ver17>
92. [https://www.microsoft.com/releasecommunications/ap...](https://www.microsoft.com/releasecommunications/api/v2/m365?responseFormat=csv) <https://www.microsoft.com/releasecommunications/api/v2/m365?responseFormat=csv>
93. New in Azure Marketplace: May 1-16, 2025 | Microsoft Community Hub <https://techcommunity.microsoft.com/blog/marketplace-blog/new-in-azure-marketplace-may-1-16-2025/4399307>
94. New in Azure Marketplace: April 22-30, 2025 <https://techcommunity.microsoft.com/blog/marketplace-blog/new-in-azure-marketplace-april-22-30-2025/4399299>
95. [PDF] Cloud Design Patterns - Download Center - Microsoft <https://download.microsoft.com/download/b/b/6/bb69622c-ab5d-4d5f-9a12-b81b952c1169/clouddesignpatternsbook-pdf.pdf>
96. Add visualizations to Reporting Services mobile reports <https://learn.microsoft.com/en-us/sql/reporting-services/mobile-reports/add-visualizations-to-reporting-services-mobile-reports?view=sql-server-ver17>
97. Tag:"advance analytics" | Microsoft Community Hub <https://techcommunity.microsoft.com/tag/advance%20analytics>
98. The Trust Fabric: Decentralized Interoperability and Economic ... <https://arxiv.org/html/2507.07901v1>
99. Fortifying the Agentic Web: A Unified Zero-Trust Architecture Against ... <https://arxiv.org/html/2508.12259v3>
100. [PDF] Towards Multi-Agent Economies: Enhancing the A2A Protocol with ... <https://arxiv.org/pdf/2507.19550>
101. A Novel Zero-Trust Identity Framework for Agentic AI - arXiv.org <https://arxiv.org/html/2505.19301v2>
102. A Gossip-Enhanced Communication Substrate for Agentic AI - arXiv <https://arxiv.org/html/2512.03285v1>
103. Securing AI Agents in Cyber-Physical Systems - arXiv <https://arxiv.org/html/2601.20184v1>
104. Sentinel Agents for Secure and Trustworthy Agentic AI in Multi-Agent ... <https://arxiv.org/html/2509.14956v1>

105. [PDF] A Unified Zero-Trust Architecture Against Logic-layer Threats - arXiv <https://arxiv.org/pdf/2508.12259>
106. Agent metadata and discoverability patterns - Microsoft Learn <https://learn.microsoft.com/en-us/entra/agent-id/identity-platform/agent-metadata-discoverability>
107. Security and Safety in the Model Context Protocol Ecosystem <https://arxiv.org/html/2512.08290v2>
108. Get Event Schema Set Definition - REST API (EventSchemaSet) <https://learn.microsoft.com/en-us/rest/api/fabric/eventscemaset/items/get-event-schema-set-definition>
109. Items - REST API (EventSchemaSet) | Microsoft Learn <https://learn.microsoft.com/en-us/rest/api/fabric/eventscemaset/items>
110. Items - List Event Schema Sets - REST API (EventSchemaSet) <https://learn.microsoft.com/en-us/rest/api/fabric/eventscemaset/items/list-event-schema-sets>
111. Items - Create Event Schema Set - REST API (EventSchemaSet) <https://learn.microsoft.com/en-us/rest/api/fabric/eventscemaset/items/create-event-schema-set>
112. Items - Update Event Schema Set - REST API (EventSchemaSet) <https://learn.microsoft.com/en-us/rest/api/fabric/eventscemaset/items/update-event-schema-set>
113. Update Event Schema Set Definition - REST API (EventSchemaSet) <https://learn.microsoft.com/en-us/rest/api/fabric/eventscemaset/items/update-event-schema-set-definition>
114. EventSchemaSet 定义 - Fabric REST APIs - Microsoft Learn <https://learn.microsoft.com/zh-cn/rest/api/fabric/articles/item-management/definitions/eventscemaset-definition>
115. Get Event Schema Set Definition - REST API (EventSchemaSet) <https://learn.microsoft.com/fr-fr/rest/api/fabric/eventscemaset/items/get-event-schema-set-definition>
116. Items - Create Event Schema Set - REST API (EventSchemaSet) <https://learn.microsoft.com/zh-cn/rest/api/fabric/eventscemaset/items/create-event-schema-set>
117. [PDF] Which LLM Multi-Agent Protocol to Choose? - arXiv.org <https://arxiv.org/pdf/2510.17149?>
118. AWCP: A Workspace Delegation Protocol for Deep-Engagement ... <https://arxiv.org/html/2602.20493v1>

119. Internet of Agents: Fundamentals, Applications, and Challenges <https://arxiv.org/html/2505.07176v2>
120. An Open-Source White-Box Agentic Benchmark Suite for AI-Native ... <https://arxiv.org/html/2601.09393v1>
121. From Prompt Injections to Protocol Exploits: Threats in LLM ... - arXiv <https://arxiv.org/html/2506.23260v1>
122. Agentic Services Computing - arXiv <https://arxiv.org/html/2509.24380v1>
123. [PDF] A Gossip-Enhanced Communication Substrate for Agentic AI <https://arxiv.org/pdf/2512.03285>
124. Advancing Multi-Agent Systems Through Model Context Protocol <https://arxiv.org/html/2504.21030v1>
125. Protecting AI conversations at Microsoft with Model Context Protocol ... <https://www.microsoft.com/insidetrack/blog/protecting-ai-conversations-at-microsoft-with-model-context-protocol-security-and-governance/>
126. Arxiv今日论文 | 2025-12-22 - 闲记算法 http://lonepatient.top/2025/12/22/arxiv_papers_2025-12-22
127. 人工智能2026_3_4 - arXiv每日学术速递 <https://www.arxivdaily.com/thread/77309>
128. (PDF) TRiSM for Agentic AI: A Review of Trust, Risk, and Security ... https://www.researchgate.net/publication/392406518_TRiSM_for_Agentic_AI_A_Review_of_Trust_Risk_and_Security_Management_in_LLM-based_Agentic_Multi-Agent_Systems
129. [PDF] Proceedings of the The First Workshop on LLM Security (LLMSEC) <https://aclanthology.org/2025.llmsec-1.pdf>
130. [PDF] Ivanti Connect Secure Attributions - 22.7R2.10 https://help.ivanti.com/ps/help/en_US/ICS/22.x/22.7R2.10/ICS%20Attributions.pdf
131. Release Notes | Red Hat Fuse | 6.3 https://docs.redhat.com/es/documentation/red_hat_fuse/6.3/epub/release_notes/index
132. Towards Multi-Agent Economies: Enhancing the A2A Protocol with ... <https://arxiv.org/html/2507.19550v1>
133. SoK: Trust-Authorization Mismatch in LLM Agent Interactions <https://arxiv.org/html/2512.06914v1>
134. The New Interoperability Paradigm: Model Context Protocol (MCP ... https://www.researchgate.net/publication/397992751_The_New_Interoperability_Paradigm_Model_Context_Protocol_MCP_APIs_and_the_Future_of_Agentic_AI
135. Computer Science Jan 2019 - arXiv <http://arxiv.org/list/cs/2019-01?skip=170&show=2000>

136. [PDF] Formalizing the Safety, Security, and Functional Properties of ... - arXiv <https://arxiv.org/pdf/2510.14133>
137. Microsoft Fabric March 2024 Update <https://blog.fabric.microsoft.com/en-us/blog/microsoft-fabric-march-2024-update/>
138. Microsoft Fabric May 2024 Update <https://blog.fabric.microsoft.com/en-us/blog/microsoft-fabric-may-2024-update>
139. What's new in Microsoft Fabric? archive <https://learn.microsoft.com/en-us/fabric/fundamentals/whats-new-archive>
140. A Tutorial on Cognitive Biases in Agentic AI-Driven 6G Autonomous ... <https://arxiv.org/html/2510.19973v1>
141. A Foundational Security Framework for Autonomous AI Agents <https://arxiv.org/html/2508.19267v1>
142. Threats in LLM-Powered AI Agents Workflows - arXiv <https://arxiv.org/html/2506.23260v2>
143. AAGATE: A NIST AI RMF-Aligned Governance Platform for Agentic AI <https://arxiv.org/html/2510.25863v1>
144. TRiSM for Agentic AI: A Review of Trust, Risk, and Security ... - arXiv <https://arxiv.org/html/2506.04133v1>
145. Toward a Safe Internet of Agents - arXiv <https://arxiv.org/html/2512.00520v1>
146. A Multimodal Framework for Preventing Prompt Injection Attacks <https://arxiv.org/html/2512.23557>
147. 100k Terms | PDF | Language Arts & Discipline | Wellness - Scribd <https://www.scribd.com/document/527844738/100k-Terms>